Lecture 6. Jointly Distributed Random Variables

Joint and marginal laws, independence, convolutions, conditional laws, order statistics, Jacobians

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6.1 Joint distribution functions

The **joint cdf** of X, Y is $F(a,b) = P(X \le a, Y \le b)$; the marginals are $F_X(a) = F(a,\infty)$ and $F_Y(b) = F(\infty, b)$, and

$$P(a_1 < X \le a_2,\ b_1 < Y \le b_2) = F(a_2,b_2) - F(a_1,b_2) - F(a_2,b_1) + F(a_1,b_1).$$

Discrete: joint pmf $p(x,y) = P(X = x, Y = y)$, $p_X(x) = \sum_y p(x,y)$. **Jointly continuous** with density f: $P((X,Y) \in C) = \iint_C f$, $f = \partial^2 F/\partial a\,\partial b$, $f_X(x) = \int f(x,y)\,dy$; likewise for n variables.

Example 6.1

$f(x,y) = 2e^{-x}e^{-2y}$, $x, y > 0$: $P(X > 1, Y < 1) = e^{-1}(1 - e^{-2}) \approx 0.318$ and

$$P(X < Y) = \int_0^\infty \int_0^y 2e^{-x}e^{-2y}\,dx\,dy = \int_0^\infty 2e^{-2y}(1 - e^{-y})\,dy = 1 - \tfrac{2}{3} = \tfrac{1}{3}.$$

Example 6.2 (A random point in a disc)

Uniform on the disc of radius R: $f_X(x) = \frac{2\sqrt{R^2 - x^2}}{\pi R^2}$, $|x| \le R$ (not uniform); the distance D to the centre has $P(D \le a) = a^2/R^2$, $f_D(a) = 2a/R^2$, $E[D] = 2R/3$.

Example 6.3 (Multinomial distribution)

n independent trials with outcome probabilities $p_1, \dots, p_r$; the counts satisfy

$$P(X_1 = n_1, \dots, X_r = n_r) = \frac{n!}{n_1! \cdots n_r!} p_1^{n_1} \cdots p_r^{n_r}, \qquad \sum n_i = n,$$

($\binom{n}{n_1,\dots,n_r}$ sequences of probability $\prod p_i^{n_i}$). Marginally $X_i \sim \mathrm{Bin}(n, p_i)$.

6.2 Independent random variables

Motivation. Independence factors the joint law into marginals.

X and Y are **independent** if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for all A, B; equivalently $F(a,b) = F_X(a)F_Y(b)$, or $p(x,y) = p_X(x)p_Y(y)$, or $f(x,y) = f_X(x)f_Y(y)$.

Proposition 6.1 (Factorization)

Jointly continuous (or discrete) X, Y are independent iff the joint density (pmf) factors as $f(x,y) = h(x)g(y)$ for all x, y.

Proof. If $f = hg$, with $C_1 = \int h$, $C_2 = \int g$: $C_1C_2 = 1$, $f_X = hC_2$, $f_Y = gC_1$, so $f_X f_Y = f$. ∎

It must hold on the whole plane: $f = 24xy$ on $\{x, y > 0, x+y < 1\}$ is not a product (non-rectangular support).

Example 6.4 (Thinning a Poisson count)

$N \sim \mathrm{Poisson}(\lambda)$ customers, each independently a man with probability p; X, Y the numbers

of men and women:

$$P(X=i,Y=j)=e^{-\lambda}\frac{\lambda^{i+j}}{(i+j)!}\binom{i+j}{i}p^iq^j=e^{-\lambda p}\frac{(\lambda p)^i}{i!}\cdot e^{-\lambda q}\frac{(\lambda q)^j}{j!}:$$

independent Poissons with means λp, λq (for fixed n they would be strongly dependent).

Example 6.5 (Meeting)

Two people arrive independently uniformly in a 30-minute window and wait 10 minutes: they meet iff $|X-Y|\le 10$, $P=1-(20/30)^2=5/9$.

Example 6.6 (Buffon's needle)

A needle of length L falls on lines $D\ge L$ apart; the centre's distance to the nearest line is $X\sim U(0,D/2)$ and the acute angle $\theta\sim U(0,\pi/2)$, independent. It crosses iff $X<\frac{L}{2}\cos\theta$:

$$P(\text{cross})=\frac{4}{\pi D}\int_0^{\pi/2}\int_0^{(L/2)\cos\theta}dx\,d\theta=\frac{2L}{\pi D}.$$

(a Monte Carlo estimate of π).

Example 6.7 (Characterizing the normal)

If X,Y are i.i.d. with differentiable density f and $f(x)f(y)=g(x^2+y^2)$, then $\frac{f'(x)}{2xf(x)}=\frac{g'(x^2+y^2)}{g(x^2+y^2)}$ for all x,y; the left side is free of y, hence both are constant: $f'(x)/f(x)=cx$, $f(x)=ke^{cx^2/2}$, $c<0$: normal with mean 0.

6.3 Sums of independent random variables

Motivation. Sums of independent variables have convolution laws; normal, Poisson and gamma families are closed under sums.

For independent X,Y with densities,

$$F_{X+Y}(a)=\int F_X(a-y)f_Y(y)\,dy,\qquad f_{X+Y}(a)=\int f_X(a-y)f_Y(y)\,dy,$$

the **convolution** of f_X and f_Y.

Example 6.8 (Sums of uniforms)

For i.i.d. $U(0,1)$, $f_{X+Y}(a)=a$ on $[0,1]$, $2-a$ on $[1,2]$; by induction $P(S_n\le x)=x^n/n!$ on $[0,1]$. With $N=\min\{n:S_n>1\}$, $P(N>n)=1/n!$, so

$$E[N]=\sum_{n\ge0}P(N>n)=\sum_{n\ge0}\frac{1}{n!}=e.$$

Proposition 6.2

For independent variables:
(a) $\text{Gamma}(s,\lambda)+\text{Gamma}(t,\lambda)\sim\text{Gamma}(s+t,\lambda)$; in particular a sum of n independent $\text{Exp}(\lambda)$ is $\text{Gamma}(n,\lambda)$ and $Z_1^2+\cdots+Z_n^2\sim\chi^2_n$.
(b) $\sum N(\mu_i,\sigma_i^2)\sim N(\sum\mu_i,\sum\sigma_i^2)$.
(c) $\text{Poisson}(\lambda_1)+\text{Poisson}(\lambda_2)\sim\text{Poisson}(\lambda_1+\lambda_2)$.
(d) $\text{Bin}(n,p)+\text{Bin}(m,p)\sim\text{Bin}(n+m,p)$.

Proof. (a) The convolution is $\frac{\lambda e^{-\lambda a}}{\Gamma(s)\Gamma(t)}\lambda^{s+t-1}\int_0^a y^{t-1}(a-y)^{s-1}\,dy = Ca^{s+t-1}e^{-\lambda a}$ ($y=ax$), a gamma density; C is forced.

(b) Reduce to $N(0,\sigma^2)+N(0,1)$ and complete the square (or use mgfs, Lecture 7).

(c) $P(X+Y=n)=\sum_k e^{-\lambda_1}\frac{\lambda_1^k}{k!}e^{-\lambda_2}\frac{\lambda_2^{n-k}}{(n-k)!}=e^{-(\lambda_1+\lambda_2)}\frac{(\lambda_1+\lambda_2)^n}{n!}$ by the binomial theorem.

(d) Count successes in $n+m$ trials, or use Vandermonde. ∎

Example 6.9

26 games won w.p. 0.4 and 18 w.p. 0.7, independently: wins have mean 23, variance 10.02, and $P(\text{wins}\ge 25)\approx 1-\Phi(1.5/\sqrt{10.02})\approx 0.32$.

6.4 Conditional distributions

Discrete: $p_{X|Y}(x\mid y)=p(x,y)/p_Y(y)$. Jointly continuous:

$$f_{X|Y}(x\mid y)=\frac{f(x,y)}{f_Y(y)},\qquad P(X\in A\mid Y=y)=\int_A f_{X|Y}(x\mid y)\,dx,$$

(from $P(x<X<x+dx\mid y<Y<y+dy)\approx f(x,y)\,dx\,dy/(f_Y(y)\,dy)$). Independence: the conditional law is free of y.

Example 6.10

(a) Independent Poissons λ_1,λ_2: given $X+Y=n$, $X\sim\text{Bin}(n,\lambda_1/(\lambda_1+\lambda_2))$. Independent $\text{Bin}(n_1,p)$, $\text{Bin}(n_2,p)$: given the sum, X is hypergeometric, free of p.
(b) Multinomial given $X_r=n_r$: multinomial with $n-n_r$ trials, probabilities $p_i/(1-p_r)$.
(c) $f(x,y)=e^{-x/y}e^{-y}/y$: $f_{X|Y}(x\mid y)=\frac{1}{y}e^{-x/y}$, so $P(X>1\mid Y=y)=e^{-1/y}$.

Example 6.11 (The bivariate normal)

(X,Y) is bivariate normal with parameters $\mu_x,\mu_y,\sigma_x,\sigma_y,\rho$ ($|\rho|<1$) if

$$f(x,y)=\frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}}\exp\Big\{-\frac{1}{2(1-\rho^2)}\Big[u^2-2\rho uv+v^2\Big]\Big\},\quad u=\frac{x-\mu_x}{\sigma_x},\ v=\frac{y-\mu_y}{\sigma_y}.$$

Completing the square in u: given $Y=y$, X is normal with

$$E[X\mid Y=y]=\mu_x+\rho\frac{\sigma_x}{\sigma_y}(y-\mu_y),\qquad \text{Var}(X\mid Y=y)=\sigma_x^2(1-\rho^2),$$

and $Y\sim N(\mu_y,\sigma_y^2)$. Independence iff $\rho=0$ (ρ is the correlation, Lecture 7).

Example 6.12 (Mixed variables: beta–binomial)

$P \sim U(0,1)$, $N \mid P = p \sim \mathrm{Bin}(n,p)$: $P(N=k) = \binom{n}{k} B(k+1, n-k+1) = \frac{1}{n+1}$, uniform on $\{0,\dots,n\}$, and $P \mid N = k \sim \mathrm{Beta}(k+1, n-k+1)$ (cf. Example 3.12).

6.5 Order statistics

For i.i.d. $X_1,\dots,X_n$ with density f, cdf F, the $n!$ assignments to the ordered values $X_{(1)} \le \cdots \le X_{(n)}$ are equally likely:

$$f_{X_{(1)},\dots,X_{(n)}}(x_1,\dots,x_n) = n!\, f(x_1)\cdots f(x_n), \qquad x_1 < \cdots < x_n.$$

$X_{(j)}$ is near x when $j-1$ observations are below, one near, $n-j$ above:

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!\,(n-j)!} F(x)^{j-1}(1-F(x))^{n-j} f(x).$$

Also $P(X_{(j)} \le x) = \sum_{k=j}^{n} \binom{n}{k} F(x)^k (1-F(x))^{n-k}$. For uniforms $X_{(j)} \sim \mathrm{Beta}(j, n-j+1)$, $E[X_{(j)}] = j/(n+1)$.

Example 6.13

(a) Three uniform points on $[0,1]$; probability no two within $d \le \frac{1}{2}$:

$$3! \int_{x_1 < x_2 - d,\ x_2 < x_3 - d} dx_1\, dx_2\, dx_3 = (1-2d)^3,$$

($y_1 = x_1$, $y_2 = x_2 - d$, $y_3 = x_3 - 2d$ maps onto the ordered simplex in $[0, 1-2d]^3$).
(b) The median of $2n+1$ uniforms has density $\frac{(2n+1)!}{n!\,n!} x^n (1-x)^n$; the range $R = X_{(n)} - X_{(1)}$ has $P(R \le a) = n \int f(x)\{F(x+a) - F(x)\}^{n-1}\, dx$, for uniforms $f_R(a) = n(n-1)a^{n-2}(1-a)$.

6.6 Functions of random variables: the Jacobian method

Motivation. The Jacobian changes variables in joint densities as in double integrals.

Theorem 6.3

Let (X_1, X_2) have density f, and let $y_1 = g_1(x_1,x_2)$, $y_2 = g_2(x_1,x_2)$ be a one-to-one differentiable map with inverse $x_i = h_i(y_1,y_2)$ and nonzero Jacobian $J(x_1,x_2) = \det(\partial g_i/\partial x_j)$. Then $Y_i = g_i(X_1,X_2)$ have joint density

$$f_{Y_1,Y_2}(y_1,y_2) = f\big(h_1(y),h_2(y)\big)\,|J(h_1(y),h_2(y))|^{-1}.$$

(The change of variables formula; also in n dimensions.)

Example 6.14 (Polar coordinates of normals)

X, Y i.i.d. $N(0,1)$, $R = \sqrt{X^2+Y^2}$, $\Theta \in [0, 2\pi)$; $(x,y) \mapsto (r,\theta)$ has Jacobian $1/r$:

$$f_{R,\Theta}(r,\theta) = \frac{1}{2\pi} e^{-r^2/2}\, r = \frac{1}{2\pi} \cdot r e^{-r^2/2}:$$

$R \perp \Theta$, Θ uniform, $R^2 \sim \mathrm{Exp}(1/2) = \chi^2_2$. Reversed (Box–Muller): $X =$

$\sqrt{-2\log U_1}\cos(2\pi U_2)$, $Y = \sqrt{-2\log U_1}\sin(2\pi U_2)$ are i.i.d. $N(0,1)$.

Example 6.15 (Gamma and beta)

Independent $X \sim \text{Gamma}(\alpha,\lambda)$, $Y \sim \text{Gamma}(\beta,\lambda)$, $U = X+Y$, $V = X/(X+Y)$: $x = uv$, $y = u(1-v)$, $|J|^{-1} = u$,

$$f_{U,V}(u,v) = \frac{\lambda e^{-\lambda u}(\lambda u)^{\alpha+\beta-1}}{\Gamma(\alpha+\beta)} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} v^{\alpha-1}(1-v)^{\beta-1}.$$

So $U \sim \text{Gamma}(\alpha+\beta,\lambda)$ and $V \sim \text{Beta}(\alpha,\beta)$ are independent, and $B(\alpha,\beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$. (E.g. for two exponential jobs in sequence, the first's share of time is uniform and independent of the total.)

Example 6.16 (Spacings of exponentials)

For i.i.d. $\text{Exp}(\lambda)$, the spacings $Y_1 = X_{(1)}$, $Y_j = X_{(j)} - X_{(j-1)}$ are independent, $Y_j \sim \text{Exp}((n-j+1)\lambda)$: the map has Jacobian 1 and $\sum_j x_{(j)} = \sum_j (n-j+1)y_j$, so $n!\lambda^n e^{-\lambda\sum x_{(j)}}$ factors (survivors restart by memorylessness). Hence $E[X_{(n)}] = \frac{1}{\lambda}\sum_{k=1}^{n}\frac{1}{k}$.

6.7 Exchangeable random variables

$X_1,\dots,X_n$ are **exchangeable** if every permutation has the same joint law. Besides i.i.d. variables, e.g. the indicators that the i-th ball drawn without replacement is white: a sequence with k whites has probability $\frac{m(m-1)\cdots(m-k+1)\,(N-m)\cdots(N-m-n+k+1)}{N(N-1)\cdots(N-n+1)}$, depending only on k; so each draw is white w.p. m/N (Example 2.1(b)). Pólya's urn also gives exchangeable draws.

6.8 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): integrating out, factorizing survival functions, and Bayes's rule for densities.

Exercise 6.5 (Ratio and product of independent variables)

Let X and Y be independent positive continuous random variables with densities f_X and f_Y.
(a) Show that $Z = X/Y$ has density $f_Z(z) = \int_0^\infty y\, f_X(zy) f_Y(y)\, dy$, $z > 0$.
(b) Show that $W = XY$ has density $f_W(w) = \int_0^\infty \frac{1}{y} f_X(w/y) f_Y(y)\, dy$, $w > 0$.
(c) Evaluate f_Z when $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(\mu)$.

Solution.

(a)

Step 1. The joint density is $f_X(x)f_Y(y)$ (Proposition 6.1); for $z > 0$

$$F_Z(z) = P(X \le zY) = \int_0^\infty \Big(\int_0^{zy} f_X(x)\,dx\Big) f_Y(y)\,dy.$$

Step 2. Substitute $x = uy$ ($dx = y\,du$) and exchange the order (Tonelli):

$$F_Z(z) = \int_0^z \Big(\int_0^\infty y\, f_X(uy) f_Y(y)\,dy\Big)\,du.$$

So the bracket is a density of Z.

(b) Likewise $F_W(w) = \int_0^\infty \int_0^{w/y} f_X(x)\,dx\, f_Y(y)\,dy$; substitute $x = u/y$ and exchange: $F_W(w) = \int_0^w \int_0^\infty \frac{1}{y} f_X(u/y) f_Y(y)\,dy\,du$.

(c) By (a), for $z > 0$,

$$f_Z(z) = \int_0^\infty y\,\lambda e^{-\lambda z y} \mu e^{-\mu y}\,dy = \frac{\lambda\mu}{(\lambda z + \mu)^2},$$

($\int_0^\infty y e^{-ay}\,dy = 1/a^2$, $a = \lambda z + \mu$). ■

Exercise 6.8 (Hazard rate of a minimum)

Let X and Y be independent positive continuous random variables with hazard rates λ_X and λ_Y, and let $W = \min(X, Y)$.
(a) Express the distribution function of W through F_X and F_Y.
(b) Show that $\lambda_W(t) = \lambda_X(t) + \lambda_Y(t)$.

Solution.

(a) $W > t$ iff $X > t$ and $Y > t$: $\bar{F}_W = \bar{F}_X \bar{F}_Y$, $F_W(t) = 1 - \{1 - F_X(t)\}\{1 - F_Y(t)\}$.

(b)

Step 1. Where f_X, f_Y are continuous, by the product rule

$$f_W(t) = -\bar{F}_W'(t) = f_X(t)\bar{F}_Y(t) + \bar{F}_X(t) f_Y(t).$$

Step 2. Divide by $\bar{F}_W(t) = \bar{F}_X(t)\bar{F}_Y(t) > 0$.

So $\min(\mathrm{Exp}(\lambda), \mathrm{Exp}(\mu)) \sim \mathrm{Exp}(\lambda + \mu)$: competing risks add. ■

Exercise 6.14 (Two geometric waiting times given their sum)

Let X and Y be independent geometric random variables with the same parameter p. Find $P(X = i \mid X + Y = n)$.

Solution.

Step 1. (Guess.) X and $X + Y$ are the times of the first and second heads of a p-coin; given the second at flip n, the first should be uniform on $1, \ldots, n-1$.

Step 2. (Verification.) With $q = 1 - p$, $n \geq 2$, $1 \leq i \leq n - 1$,

$$P(X = i,\, Y = n - i) = q^{i-1}p \cdot q^{n-i-1}p = p^2 q^{n-2},$$

free of i; so $P(X + Y = n) = (n-1)p^2 q^{n-2}$ and

$$P(X = i \mid X + Y = n) = \frac{1}{n-1}, \qquad i = 1, \ldots, n-1.$$

■

Exercise 6.21 (Poisson counts with a gamma rate)

The moisture W on a day is $\mathrm{Gamma}(t, \lambda)$, and given $W = w$ the number N of accidents that day is $\mathrm{Poisson}(w)$. Show that given $N = n$, W is $\mathrm{Gamma}(t + n, \lambda + 1)$.

Solution.

Step 1. By Bayes's formula for the mixed pair (as in Example 6.12),

$$f_{W|N}(w \mid n) = \frac{P(N = n \mid W = w) f_W(w)}{P(N = n)}, \qquad P(N = n) = \int_0^\infty P(N = n \mid W = w) f_W(w)\, dw.$$

Step 2. For $w > 0$,

$$P(N = n \mid W = w) f_W(w) = e^{-w} \frac{w^n}{n!} \cdot \frac{\lambda e^{-\lambda w} (\lambda w)^{t-1}}{\Gamma(t)} = C_n\, e^{-(\lambda+1)w} w^{t+n-1},$$

with $C_n = \lambda^t / \{n!\, \Gamma(t)\}$ free of w.

Step 3. So $f_{W|N}(w \mid n) \propto e^{-(\lambda+1)w} w^{t+n-1}$, the Gamma$(t + n, \lambda + 1)$ kernel; proportional densities are equal. In particular $E[W \mid N = n] = (t + n)/(\lambda + 1)$. ■